Systemic Vascular Resistance

Systemic vascular resistance (SVR) is the resistance to flow in the peripheral vascular tree that must be overcome for blood to circulate. The normal range is about 900-1200 dynes*sec/cm^{-5}. The major determinant of systemic vascular resistance is arteriolar tone. Dilation or constriction of these vessels results from stimulation or inhibition by a wide variety of substances, mostly released locally in response to changes in local tissue metabolism. For example, decreased tissue oxygen and increased tissue carbon dioxide levels lead to arteriolar vasodilation whereas catecholamines such as epinephrine and norepinephrine can result in vasoconstriction. The components critical to determining the SVR are the mean arterial pressure (MAP), central venous pressure (CVP), and the cardiac output (CO). The result of the SVR formula is multiplied by a coefficient of 80 to convert it into the units we commonly use to measure it. The MAP and CVP are both calculated in mmHg and the cardiac output is in the form of liters per minute (L/min). The formula for systemic vascular resistance is:

$$SVR = \frac{MAP - CVP}{CO} \times 80$$

In a patient with a mean arterial pressure of 85, a central venous pressure of 12, and a cardiac output of 5.5 L/min, we would set the problem up as:

$$SVR = \frac{85 - 12}{5.5} \times 80 = \frac{73}{5.5} \times 80 = 13.27 \times 80$$

$$= 1061.6 \text{ dyne*secs/cm}^{-5}$$

Calculate the SVR for a patient with a MAP of 69, a CVP of 8, and a cardiac output of 4.0 L/min.

$$SVR = \frac{MAP - CVP}{CO} \times 80$$

$$= \frac{69 - 8}{4.0} \times 80$$

$$= \frac{61}{4} \times 80$$

$$= 1220 \text{ dyne*secs/cm}^{-5}$$

Calculate the systemic vascular resistance for patients in whom the:

1. MAP = 84 mmHg, CVP = 15 mmHg, and CO = 5.3 L/min
2. MAP = 101 mmHg, CVP = 3 mmHg, and CO = 9.8 L/min
3. MAP = 62 mmHg, CVP = 6 mmHg, and CO = 3.1 L/min
4. MAP = 75 mmHg, CVP = 9 mmHg, and CO = 4.6 L/min
5. MAP = 81 mmHg, CVP = 12 mmHg, and CO = 1.8 L/min
6. MAP = 105 mmHg, CVP = 4 mmHg, and CO = 15.3 L/min
7. MAP = 90 mmHg, CVP = 8 mmHg, and CO = 9.2 L/min
8. MAP = 83 mmHg, CVP = 16 mmHg, and CO = 3.7 L/min
9. MAP = 74 mmHg, CVP = 5 mmHg, and CO = 6.5 L/min
10. MAP = 63 mmHg, CVP = 10 mmHg, and CO = 5.8 L/min
11. MAP = 95 mmHg, CVP = 11 mmHg, and CO = 4.9 L/min
12. MAP = 44 mmHg, CVP = 2 mmHg, and CO = 8.3 L/min
13. MAP = 99 mmHg, CVP = 13 mmHg, and CO = 7.8 L/min
14. MAP = 86 mmHg, CVP = 8 mmHg, and CO = 5.0 L/min
15. MAP = 65 mmHg, CVP = 5 mmHg, and CO = 6.1 L/min
16. MAP = 70 mmHg, CVP = 11 mmHg, and CO = 7.8 L/min
17. MAP = 51 mmHg, CVP = 9 mmHg, and CO = 9.1 L/min
18. MAP = 59 mmHg, CVP = 14 mmHg, and CO = 3.4 L/min
19. MAP = 115 mmHg, CVP = 10 mmHg, and CO = 4.8 L/min
20. MAP = 35 mmHg, CVP = 1 mmHg, and CO = 7.7 L/min
21. MAP = 79 mmHg, CVP = 15 mmHg, and CO = 6.9 L/min
22. MAP = 64 mmHg, CVP = 3 mmHg, and CO = 5.5 L/min
23. MAP = 55 mmHg, CVP = 7 mmHg, and CO = 5.1 L/min
24. MAP = 88 mmHg, CVP = 4 mmHg, and CO = 4.3 L/min
25. MAP = 110 mmHg, CVP = 6 mmHg, and CO = 5.8 L/min

Systemic Vascular Resistance Calculation Answers

(answers are in dyne*secs/cm^{-5})

1. 1042	6. 528	11. 1371	16. 605	21. 742
2. 800	7. 713	12. 405	17. 369	22. 887
3. 1445	8. 1449	13. 882	18. 1059	23. 753
4. 1148	9. 849	14. 1248	19. 1750	24. 1563
5. 3067	10. 731	15. 787	20. 353	25. 1434